

Enterprise Resource Planning (ERP) System Requirements Document

1. Project Overview

1.1 Project Name

Enterprise Resource Planning (ERP) System

1.2 Project Description

The ERP system is a microservices based enterprise resource planning solution designed to manage business operations efficiently. It will be developed using Spring Boot for backend services and React.js for the frontend. The system will integrate different business modules, including Finance, POS, CRM, Purchases, Inventory, Sales, Fixed Assets, Banks, and HR.

1.3 Technology Stack

- Backend: Spring Boot (Microservices architecture)
- Frontend: React.js
- Database: PostgreSQL / mySql / MongoDB
- API Communication: RESTful APIs / GraphQL
- Authentication & Security: JWT
- Messaging & Event Streaming: Kafka.

2. System Architecture

The ERP system will be designed using microservices architecture, ensuring modularity, scalability, and ease of integration. Each service will have its own database to maintain data consistency and autonomy.

2.1 Microservices List

Each microservice will be responsible for a specific domain:

1. Finance Service
2. Point of Sale (POS) Service
3. Customer Relationship Management (CRM) Service
4. Purchases Service
5. Inventory Management Service
6. Sales Service
7. Fixed Assets Service
8. Banks Service
9. Human Resources (HR) Service

2.2 Communication Between Services

API Gateway: Acts as a single entry point for clients.

Service Discovery: Using Eureka or Consul.

Interservice Communication: Using REST APIs, gRPC, or Kafka/RabbitMQ for eventdriven communication.

3. Functional Requirements

Below are the key functionalities for each module:

3.1 Finance Service

- Manage financial accounts, transactions, and general ledger.
- Handle accounts payable & receivable.
- Generate financial reports (balance sheet, profit & loss).
- Budgeting and forecasting.
- Automated journal entries and ledgers.
- Simplified tax filing and calculations.

3.2 Point of Sale (POS) Service

- Process sales transactions in realtime.
- Manage cash register and multiple payment methods.
- Integration with inventory for stock updates.
- Sales analytics and reporting.

3.3 Customer Relationship Management (CRM) Service

- Manage customer database and interactions.
- Track sales leads, opportunities, and deals.
- Automate followups and email campaigns.
- Customer support ticketing system.

3.4 Purchases Service

- Manage supplier information.
- Purchase order creation and approval.
- Track order status and delivery.
- Generate purchase reports and analytics.

3.5 Inventory Management Service

- Manage product catalog and stock levels.
- Automated stock reordering based on demand.
- Warehouse management with multilocation support.
- Inventory tracking and reporting.
- Detailed reports on all items.
- Tracking inventory in all branches.
- Alerts for low ingredient levels.
- Managing supplier data and purchase orders.

3.6 Sales Service

- Manage sales orders and invoices.
- Customer order tracking and history.
- Pricing and discount management.
- Sales performance reporting.

3.7 Fixed Assets Service

- Maintain records of company assets.
- Depreciation calculation and tracking.
- Asset transfer and disposal management.
- Generate fixed asset reports.
- Generate reports and analytics.

3.8 Banks Service

- Manage company bank accounts.
- Bank transactions reconciliation.
- Monitor cash flow and fund transfers.
- Bank statements and audit reports.

3.9 Human Resources (HR) Service

- Employee records and payroll management.
- Leave and attendance tracking.
- Performance evaluations and appraisals.
- Employee benefits and taxation management.
- Create new user for the system.
- Determine user promotions.
- User should submit personal leave requests, business leave, overtime.